

KHALIL KRIFI

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EDUCATION

Computer Science Engineering, ENSI, University of Manouba
Engineering Preparatory Studies, IPEIN, Nabeul

Sep 2023 \ Jun 2026
Sep 2021 \ Jun 2023

EXPERIENCE

Graduation Internship: Natural Language Query Interface for logs and Metrics
Cognira, Tunis, Tunisia

Feb 2026 \ Jun 2026

- Built an **event-driven 3-service platform** that converts natural language into **OpenSearch DSL** and **Prometheus PromQL**: a **Scala** backend (Tapir + Pekko HTTP + jOOQ), a **Python** AI worker (FastAPI + LangGraph), and a **Next.js 14** frontend, decoupled over **Apache Kafka**.
- Engineered a **schema-aware LLM pipeline** on a self-hosted **vLLM** model (Qwen2.5-Coder-7B-AWQ) with a **Lang-Graph self-correction loop** that validates and repairs generated queries, rejecting hallucinated fields.
- Hardened the platform with a **semantic cache** (Redis + 384-dim embeddings, cosine ≥ 0.95) to cut redundant LLM calls, a **multi-stage DSL security guard** enforcing a field whitelist, and **multi-tenant RBAC**, **TOTP MFA**, **AES-256**-encrypted API keys, and a BFF proxy storing JWTs in httpOnly cookies.
- Deployed to **Kubernetes** via **Kustomize** and **ArgoCD GitOps** with **External Secrets**, and validated scalability through **load, spike, and performance testing** (Locust, K6) with **Prometheus/Grafana** observability.

Software Engineer Intern: Fleet Management System (repo)
Omnilink, Tunis, Tunisia (remote)

Jul 2025 \ Sep 2025

- Designed and shipped a **fleet-management platform with 10 Spring Boot microservices**, containerized with Docker and orchestrated via **docker-compose**, enabling modular deployment and independent scaling.
- Delivered **40+ GraphQL APIs** (Apollo and Netflix DGS) exposing vehicle, driver, and trip data to the frontend.
- Enabled **real-time updates over WebSockets** for vehicle status, live location streams, and alerting.
- Integrated **OAuth 2.0 / OpenID Connect** with Keycloak for authentication and role-based access control.

AI Research / Engineering Intern: Falsified Signature Detection (repo)
Laval University, Quebec City, Canada (remote)

Feb 2025 \ May 2025

- Led the design and training of **8+ binary classifiers** using **pretrained backbones via transfer learning**.
- Optimized performance through 50+ Optuna hyperparameter trials, achieving **94% accuracy** and **93% F1** on test.
- Augmented data with Albumentations (rotations, scaling, noise, contrast) to **improve generalization**.
- Utilized **XAI techniques** (SHAP, LIME, Grad-CAM) to generate visual explanations for stakeholders.
- Built an **8-service microservice architecture** AI inference (Python/PyTorch), backend APIs (Spring Boot + Spring Cloud), and frontend (Angular) and **containerized** all components with Docker for reproducible deployment.

Software Engineer Intern: Enterprise Content Management (repo)
BFI Group, Tunis, Tunisia (on-site)

Jun 2024 \ Aug 2024

- Built document management tool for files up to **500 MB**, with directory hierarchy and versioning.
- Integrated **Elasticsearch** for search and indexing across **20 000+ documents**, achieving sub-**300 ms** avg retrieval.
- Enforced **JWT-based authentication** and role-based access control with fine-grained document permissions.

PROJECTS

Etude.AI: Microservice Study Platform Scalable study assistant built with a Java and Python microservice.

- Created a **2-microservice system**: **Spring Boot** using reactive **WebClient** and **FastAPI** service.
- Integrated a **circuit breaker** to handle downstream failures and timeouts with 5 trials.
- Applied **rate limiting** (Token Bucket) using **Bucket4j** for **Spring Boot** and **SlowAPI** for **FastAPI** throttling.
- Containerized services with **Docker** and deployed on **Kubernetes** with autoscaling and health checks.
- Monitored services with **Prometheus** metrics, **Grafana** dashboards, and **Sentry** for error tracking.

SKILLS

Programming Languages: Java, TypeScript/JavaScript, Python, C/C++, SQL, GraphQL

Technologies: Git, Docker, Kubernetes, Spring Boot, Angular, React, PostgreSQL, Spark, Hadoop, HBase, Power BI

Languages: Arabic (Native), French (Fluent), English (Fluent) **Interests:** Chess, Piano